# Prince **Asumadu**

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## Research Interests

- Scientific Computing
- Numerical Linear Algebra
- Communication-Avoiding Algorithms
- Parallel & High-Performance Computing
- Krylov Subspace & Eigenproblem Methods
- Graph Algorithms & Analysis

## Education

**MPhil** **Scientific Computing and Industrial Modelling** Jan 2025 – Dec 2026 (expected)
National Institute for Mathematical Sciences (NIMS-KNUST), Ghana
- **Supervisor:** Dr. Richard Owusu

**BSc** **Mathematics** Sep 2019 – Nov 2023
Kwame Nkrumah University of Science and Technology (KNUST), Ghana

## Experience

### Research

**A Communication-Avoiding Parallel Spectral Clustering Pipeline** Jan 2026 – Present
*National Institute for Mathematical Sciences,* ***THESIS***

- Developing a spectral clustering pipeline for graph Laplacians using a communication-avoiding Krylov method (CA-Lanczos kernel) to reduce synchronizations and global reductions.
- Implemented classical and s-step communication-avoiding Lanczos eigensolvers in C with MPI, validated against SciPy's `eigsh`, and cut global reductions by about 4.6-fold at s = 8 while preserving eigenvalue accuracy.

**Research Assistant** Jan 2024 – Aug 2024
National Institute for Mathematical Sciences Kumasi

- Ran computational experiments on the institute's cluster in support of ongoing research projects, handling job submission and the collection of results.
- Tested code for ongoing research projects at the institute, checking implementations and reporting issues to the researchers responsible.

### Projects

**Optimal Task Allocation using Modified Primal Simplex for KNUST Cleaning Staff** Jan 2024 – May 2024
*National Institute for Mathematical Sciences* Project

- Formulated the allocation of tasks to employees as a linear programming problem, then implemented a modified primal simplex method to solve the resulting model.
- Compared the modified implementation against the classical primal simplex and the Hungarian algorithm on solution accuracy and runtime, and used the resulting allocation to produce a working duty roster for the cleaning staff.

**Dynamic Evacuation Plan Model for Aboagye Menyah Complex at KNUST**, Apr 2023 – Aug 2023
*Department of Mathematics, KNUST* Project
*Supervisor: Prof. Peter Amoako-Yirenkyi*

- Represented the complex as a network graph and developed a dynamic shortest-path model that takes the hazard's position on the network as input, recomputing evacuation routes as that position changes.
- Simulated evacuation under a range of hazard scenarios and specified a protocol for incorporating the model into a building's emergency and security system.

## Teaching

**Python Tutor** Oct 2025 – Nov 2025
Ghana Computational Biosciences Network Kumasi

- Designed and delivered a two-month introductory Python programming bootcamp for 20–30 members of the network, writing all course materials and exercises.
- Assigned weekly exercises following each session and supervised a final project, through which participants applied the material to their own work.

**Teaching Assistant** Jan 2024 – Dec 2024
Department of Mathematics (KNUST) Kumasi

- Tutored undergraduate courses in differential equations and numerical analysis across a full academic year, working through problem sets with students and covering material from lectures.
- Instructed and graded laboratory sections on parallel programming with OpenMP and MPI in C, supporting students through the practical exercises and assessing their submitted work.

**Substitute Teacher**, Aug 2020
Kumasi Academy Kumasi

- Instructed students on elective mathematics in online sessions.

## Professional

**Part-time Database Developer** Sep 2024 – Dec 2024
Adepts Ghana Kumasi

- Designed and built relational database schemas as part of the database team on two client systems: a hospital management system and a complaints management system for an insurance company.
- Maintained the databases supporting these systems in production, handling schema changes and data issues as they arose over the course of the engagement.

**Intern, IT Department**, Oct 2022 – Dec 2022
Ghana Ports and Harbours Authority Tema

- Diagnosed and resolved hardware and software faults on staff workstations.
- Installed, configured, and monitored LAN, Wi-Fi, and fiber-optic links across the site.

**Intern, Research Department**, Oct 2021 – Nov 2021
Manhyia Hospital Kumasi

- Compiled and analyzed outpatient department records daily, producing the summary statistics used in the department's routine reporting.
- Designed and administered a staff satisfaction study across the hospital, identified inadequate provision of COVID-19 safety equipment as the leading driver of reported dissatisfaction, and presented the findings.

## Publications and Presentations

**A Communication-Avoiding Parallel Spectral Clustering Pipeline** *In preparation*
P. Asumadu, E. S. Menkah, P. Amoako-Yirenkyi, R. Owusu
*Manuscript — In preparation for submission*

**A Communication-Avoiding Parallel Spectral Clustering Pipeline** Mar 2026
*MPhil proposal seminar* — NIMS-KNUST, Kumasi, Ghana

**Real-Time Evacuation Planning** Sep 2025
*Poster* — KNUST Scientific Conference, Kumasi, Ghana

**Task Allocation Made Easy** Dec 2024
P. Asumadu, O. Inusah
*Poster* — DS-Change Workshop: Uniting Academia & Industry, KNUST, Ghana

## Awards and Scholarships

**SCIM Scholarship** 2025 – 2026
Center for Scientific & Technical Computing NIMS-KNUST

## Skills

**Programming:** Python, C/C++, R, SQL

**HPC:** MPI, OpenMP, CUDA, OpenPBS, SLEPc, SciPy, Make, CMake

**ML:** Scikit-learn, NumPy, Pandas, Altair, Seaborn

**Other:** Git / GitHub, MATLAB, LaTeX, Shell scripting (Bash)

## Workshops and Conferences

Indaba X: Ghana Data Science Summit, Ghana Oct 2024
Deep Learning Indaba, Accra, Ghana Sep 2023
High Performance Computing for Sustainable Development, ICTP, *Virtual* Apr 2023